Reece Shiraishi

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EDUCATION

Cornell University, College of Engineering (GPA: 3.8)
B.S. Electrical and Computer Engineering, Minors in Computer Science & Artificial Intelligence Expected: May 2027

- **Relevant Courses:** Robotics, Embedded Systems, Computer Architecture, Linear Algebra, Ethics of AI, Foundations of ML, Digital Logic, Computer Architecture, Microelectronics
- **Achievements:** National Merit Semifinalist | Presidential Scholar Nominee | Eagle Scout | IEEE-HKN

SKILLS

Programming Languages: Python, Java, C, C++, Typescript, Verilog, Quartus FPGA tools, Bash scripting, R, Arduino, Assembly (ISAs: RISC-V, ARM)
Tools, Libraries, Frameworks: Git/GitHub, ROS/RViz, Arduino, NumPy, PyTorch, Pandas, Scikit-learn, SLURM cluster, CMake, I2C, LTspice, NoSQL databases, vector databases, LTSpice, oscilloscopes
Languages: English (native), Spanish (full professional fluency)

EXPERIENCE

ML Engineer Intern | SRC, Inc; Syracuse, NY *May 2026 – Aug. 2026*

- Engineered and maintained Python pipelines for EM signal challenges in Agile development team
- Trained and fine-tuned image-recognition model on spectrograms to 93.2% mAP50 accuracy
- Developed Python bounding box validator using FFT, finding 20% of files in 1200-file 1TB dataset is incorrect

Software Engineering Intern | Fermi National Accelerator Laboratory; Batavia, IL *Jun. 2025 – Aug. 2025*

- Developed Python workflow to reformat data (YAML to LaTeX/Markdown) in pair programming environment; awarded academic citation. System actively used by 7-person research team
- Trained PyTorch neural network and associated Matplotlib plots to classify particles with 70% accuracy

Full Stack Developer Intern | Wabi Skincare: Hoboken, NJ *Jan. 2025 – May 2025*

- Architected scalable user engagement features and database interfacing in fast-paced early startup environment
- Implemented storage of skincare data in NoSQL cloud database, the team's most critical feature for beta release
- Resolved bug where website tabs failed to render using asynchronous database retrieval

Research Assistant – Prof. Cafer Avci | Cornell University; Ithaca, NY *Jan. 2025 – May 2025*

- Applied GeoJSON Python packages and OpenStreetMap APIs to improve traffic flow in urban areas
- Consolidated the fragmented Python analysis pipeline, increasing the pipeline's runtime speed by 3x

PROJECTS

CAST Autograd (CNet 2.0) (C++, CMake) *Aug. 2025 – present*

- Designed deep learning framework, including full documentation and handmade SGD optimizer, from scratch
- Reorganized and optimized CNet's codebase, reducing SGD optimizer code from 300 to 10 lines
- Leveraged Github Workflows CI/CD for automated testing, removing need to manually run 10+ unit tests

Theft Protection IoT Device (Embedded C) *Apr. 2025 – May 2025*

- Developed bare-metal security device and accompanying drivers for ARM Cortex FRDM-KL46Z MCU with 32KB RAM
- Added concurrency with locks, 4 original libraries, custom round-robin scheduler, I2C interface
- Integrated passive buzzers, accelerometers, onboard LEDs, and Bluetooth modules, detecting thieves on 5/5 trials

5-Stage Pipelined Processor (SystemVerilog) *Sep. 2026 – Oct. 2026*

- Implemented processor, control unit, cache executing 27 RISC-V instructions (arithmetic, branch, jump, load/store)
- Integrated original iterative multiplier, stall logic, squashing, Python/shell/Makefile testing suite
- Augmented pipeline with bypassing, decreasing average CPI from 3.6 to 2.9 (23% improvement)

StepperFX (Java) *Jun. 2025– May 2026*

- Engineered enterprise-grade GUI application, implementing cipher algorithm to process 100M characters in <15s
- Implemented Model-View-Controller architecture, safe multithreading, and 83 automated tests